Dashain/Tihar Vacation Assignment for XI (PHYSICS-MECHANICS)

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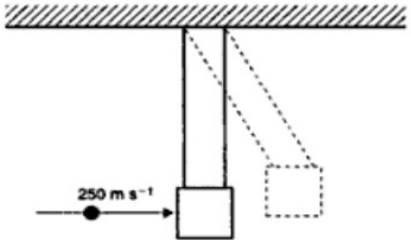
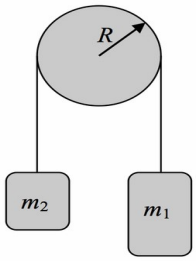
1. A bullet of mass 15 g is fired horizontally from a gun with a velocity of 250 ms^{-1}. It hits, and becomes embedded in a block of wood of mass 3000 g, which is freely suspended by long string as shown in Fig. Air resistance is to be neglected.



a) Calculate the magnitude of the momentum of the bullet as it leaves the gun.
b) Calculate the magnitude of the initial velocity of the wooden block and bullet after impact.
c) Use your answer to (b) to calculate the kinetic energy of the wooden block and embedded bullet immediately after the impact.
d) Hence, calculate the maximum height above the equilibrium position to which the wooden block, with the embedded bullet, rises after impact. (use g = 10 ms^{-2})

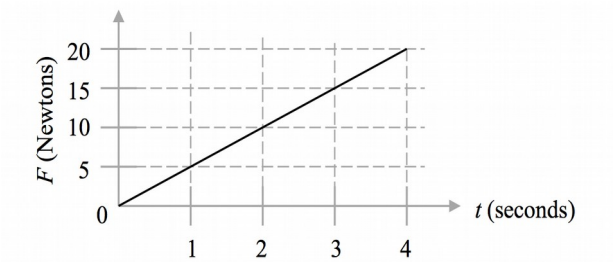
2.



Two masses are hung are connected by a light cord and hung from a frictionless pulley of negligible mass as shown. Mass m_1=3.00 kg, and mass m_2=2.00 kg. When the two masses are released from rest, what's the resulting acceleration of the two masses.

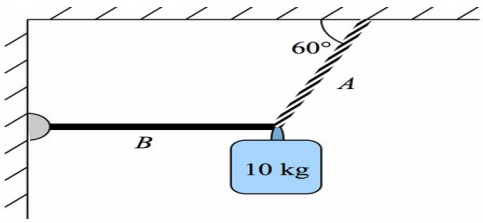
3. A 0.50 kg object moves along the x-axis according to the function $x = 4t^3+2t-1$, where x is in meters and t is in seconds. What is the magnitude of the net force acting on the object at time t=2.0 s?

4.



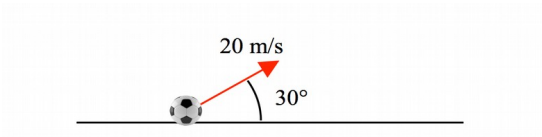
A block is pushed along a horizontal, frictionless surface, with a horizontal force that varies as a function of time as shown in the graph. At time t=3 seconds, the acceleration of the block is 5.0 m/s^2. What is the mass of the block ?

5.



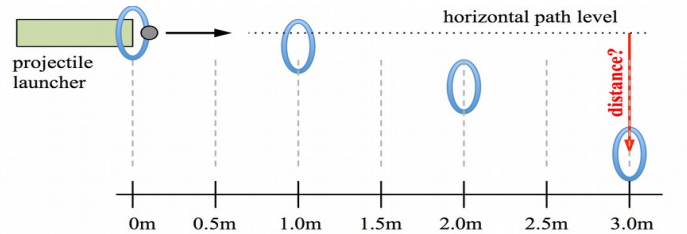
A mass of 10 kg is suspended from a cable A and a light, rigid, horizontal bar B that is free to rotate, as shown. What is the approximate tension, in cable A ?

6.



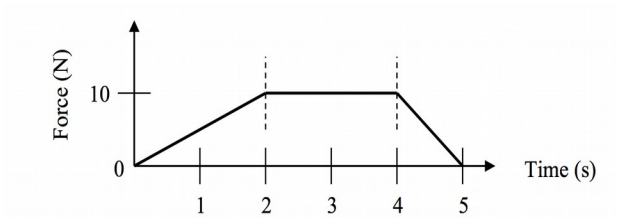
A soccer ball is kicked to give it an initial velocity of 20 m/s at 30^0 relative to the ground, as shown. What is the maximum height reached by the ball ?

7.



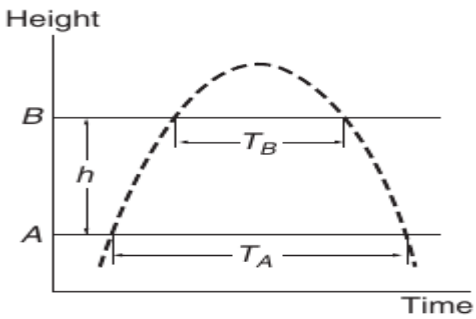
A physics teacher wants to prepare a demonstration on projectile motion for her students. A launcher, placed at the top of a building, will fire a ball horizontally, and the ball will pass through a series of elevated rings that have been set up as shown above. The ball is fired with an initial horizontal velocity of 2.0 m/s; air friction is negligible. At what distance below the horizontal path level should the fourth ring be placed if the ball is to pass through it?

8.



An object of mass m = 2.0 kg experiences a force according to the Force vs time graph shown here. For the time interval shown, what is the total change in momentum of the object?

9.

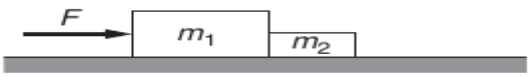


The acceleration of gravity can be measured by projecting a body upward and measuring the time it takes to pass two given points in both directions.
Show that if the time the body takes to pass a horizontal line A in both directions is T_A, and the time to go by a second line B in both directions is T_B, then, assuming that the acceleration is constant, it magnitude is

$$g = \frac{8h}{T_A{}^2 - T_B{}^2},$$

where h is the height of line B above line A.

10.



Two blocks m_1 and m_2 are in contact on a horizontal table. A horizontal force is applied to one of the blocks, as shown in the figure. If m_1=2 kg, m_2=1 kg and F=3N, find the force of contact between the two blocks.